



DEALER BEST PRACTICE SERIES

WIP Parts Storage Boxes
and Lifting Tool

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

WIP Parts Storage Boxes and Lifting Tool.....0

1.0 Introduction.....1

2.0 Best Practice Description.....1

3.0 Implementation Steps.....3

4.0 Benefits.....3

5.0 Resources Required.....3

6.0 Supporting Attachments / References4

7.0 Related Best Practices9

8.0 Acknowledgements.....9

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1.0 Introduction

Trakindo's Component Rebuild Centers (CRCs) at all times have Work in Process (WIP). WIP when not stored and arranged well can affect the centers efficiency, cleanliness (Contamination Control), quality, and safety. To assist in the storage and efficient arrangement of the WIP, Trakindo has implemented a standard WIP Parts Storage Box (yellow box) to keep parts or small components organized and safe in while waiting to complete a job.

In some CRCs with a high volume of work orders, many WIP Parts Storage Boxes may be required. These boxes are usually placed on the floor and take up a large amount of space. Currently the boxes are placed only one high and not stacked, decreasing floor space utilization.

2.0 Best Practice Description

To increase available floor space and make the workshop area more efficient, Trakindo recommends that the boxes be stacked two high using a bench (rack). Currently they do not have the capability in the workshop to use forklifts to lift the boxes.

In order to solve the inefficiency with the single height stacking practice, Trakindo has developed a lifting tool to stack the boxes on a bench and increase floor space, in turn increasing workshop efficiency.



Figure 1 and 2: Trakindo's WIP Parts Storage Box (yellow box) on CRC floor before implementation of bench (rack).

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WIP Parts Storage Boxes and Lifting Tool

DATE
2/19/2015

CHG
NO
00

NUMBER
0911-4.03-1230



Figure 3: Trakindo's WIP Parts Storage Box (yellow box) on CRC floor after implementation of bench (rack).

The lifting tool is fabricated from standard 'U' beam and box section steel with a thickness of +/- 5cm (see picture in Section 6.0). There are two separate models of fabricated steel, with an additional chain that will lift the box safely and easily, and not damage the box.

The tool is designed to make it easy and safe for double stacking the boxes, resulting in additional space and cleanliness within the shop. This is especially useful when a shop does not have sufficient space to use a forklift.

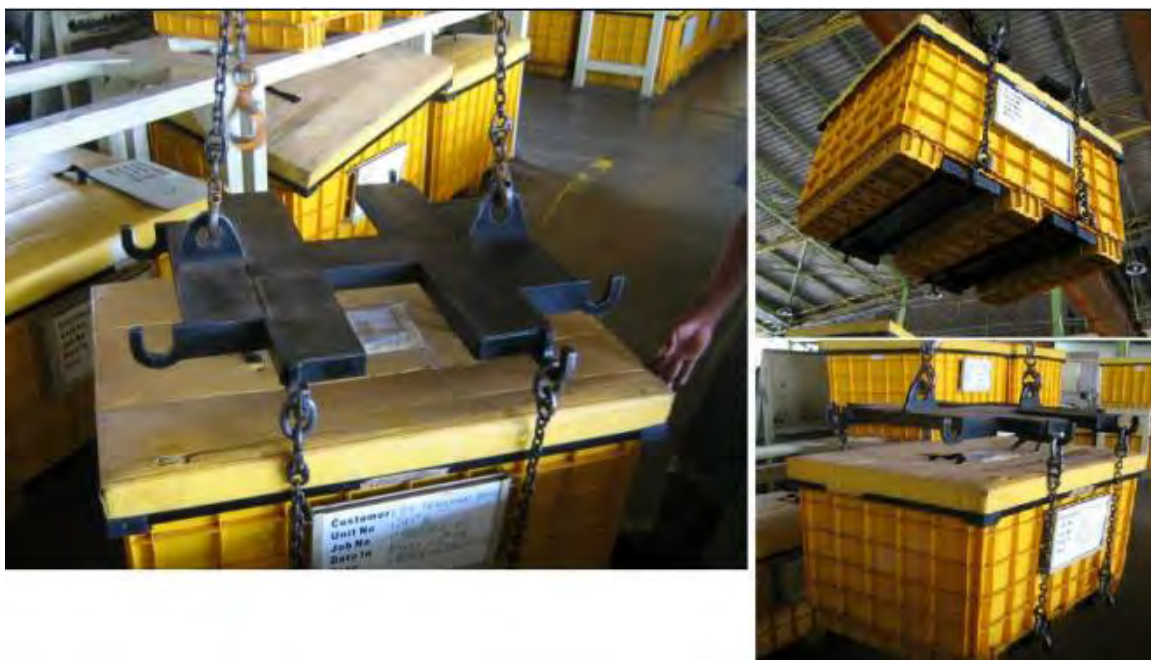


Figure 4, 5, and 6: Trakindo's WIP Parts Storage Box (yellow box) with fabricated lifting tool.

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DATE
2/19/2015

CHG
NO
00

NUMBER
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3.0 Implementation Steps

- Fabricate the tool.
- Determine area in shop to be designated for rack.
- Fabricate rack - if not currently available.
- Assure the area has adequate space and access for a crane to move the boxes.
- Assure crane is available (jip crane or overhead crane) to move the boxes to and from rack.
- Implement the layout changes into the shop.
 - o Suggest the racks are staged in the assembly bay.
- Promote change feedback as installation may create new problems.
- Train employees to use the new procedures/equipment.
- Review process/equipment performance once established.

4.0 Benefits

Safety

- Well designed/organized storage rack will improve cleaning efforts and safety.

Organization

- Yellow Boxes are consistently staged in specific areas for each process step.

Quality

- More effectively placed WIP simplifies staging and the cleaning process.
- Promotes high technician professionalism.
- Meets cleanliness (contamination control) requirements.

5.0 Resources Required

Fabrication and Facility

- Steel standard 'U' beam (5mm).
- Reorganized the work area to allow for stacking of incoming and outgoing boxes.

People

- Provide training on use of lifting device.
- Establish the new process, lay out, and equipment improvement through the team.
- Remove replaced items from operation to sustain the gains.

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WIP Parts Storage Boxes and Lifting Tool	DATE 2/19/2015	CHG NO 00	NUMBER 0911-4.03-1230
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6.0 Supporting Attachments / References

Pictures

- Note all dimensions are specific to the Trakindo Storage Boxes.
- Dimensions should be customized for any replication of this best practice to allow the dealer/customer to “fit” to their boxes.
- Tooling capacity 100 kg.

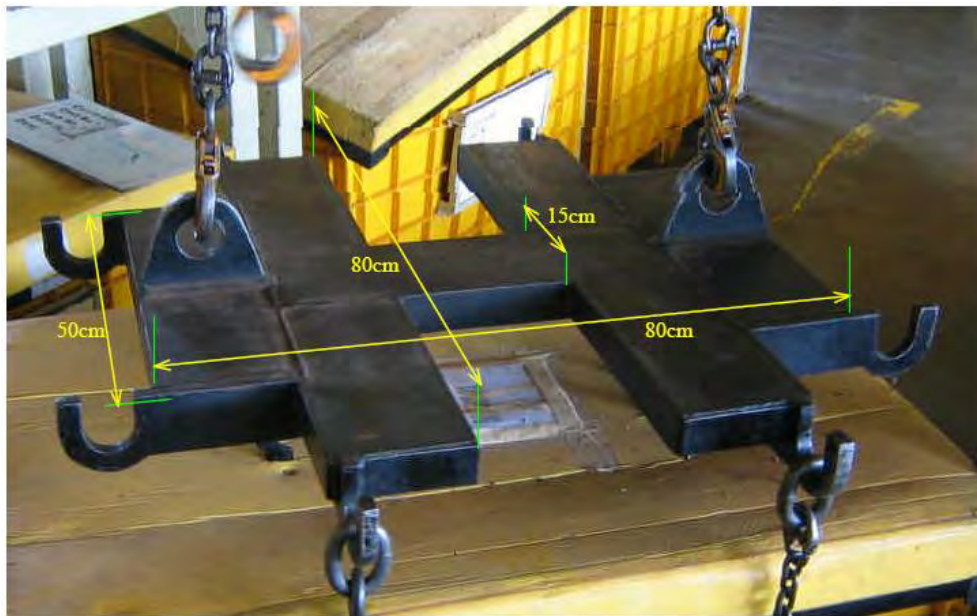


Figure 7: Top portion of lifting tool.

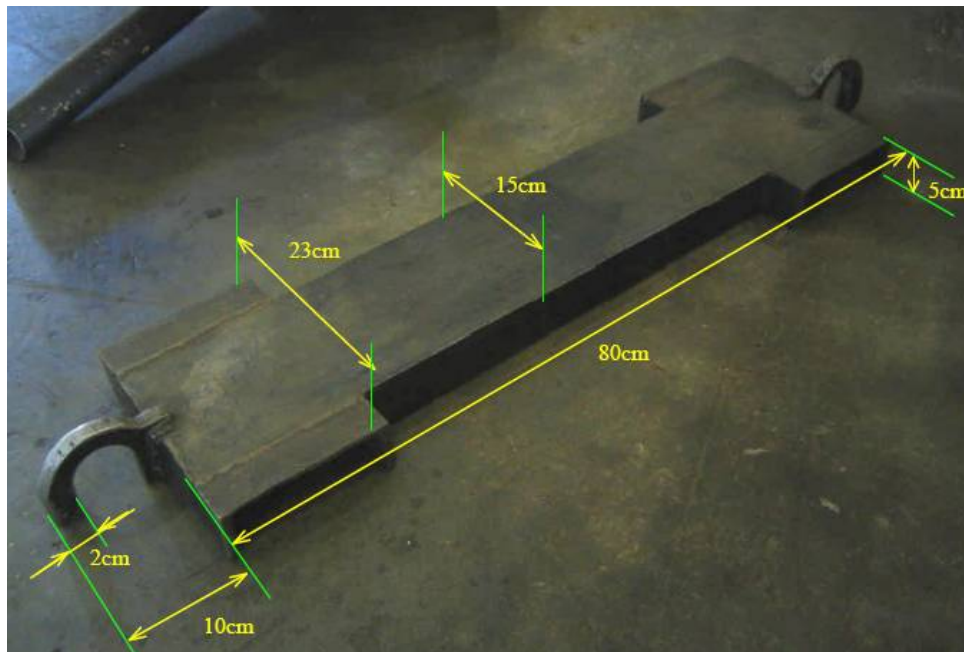


Figure 8: Bottom portion of lifting tool (2 pieces – see Figure 5).

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WIP Parts Storage Boxes and Lifting Tool

DATE
2/19/2015

CHG
NO
00

NUMBER
0911-4.03-1230

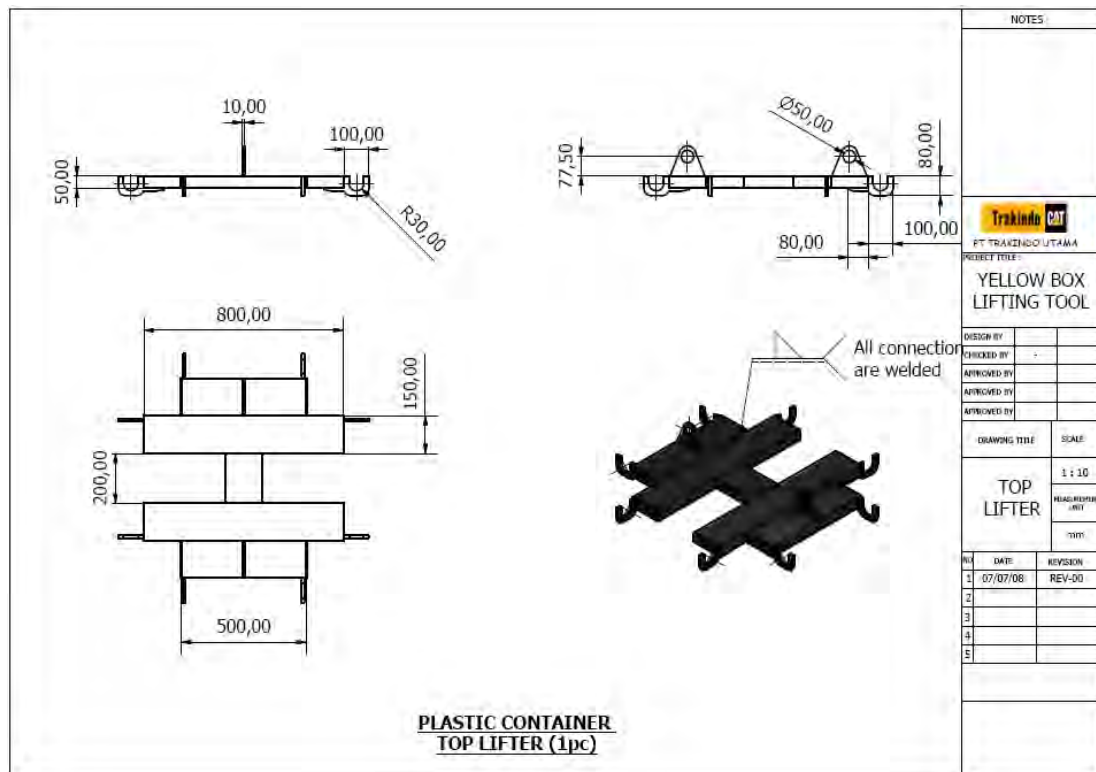


Figure 9: Drawing of top portion of lifting tool.

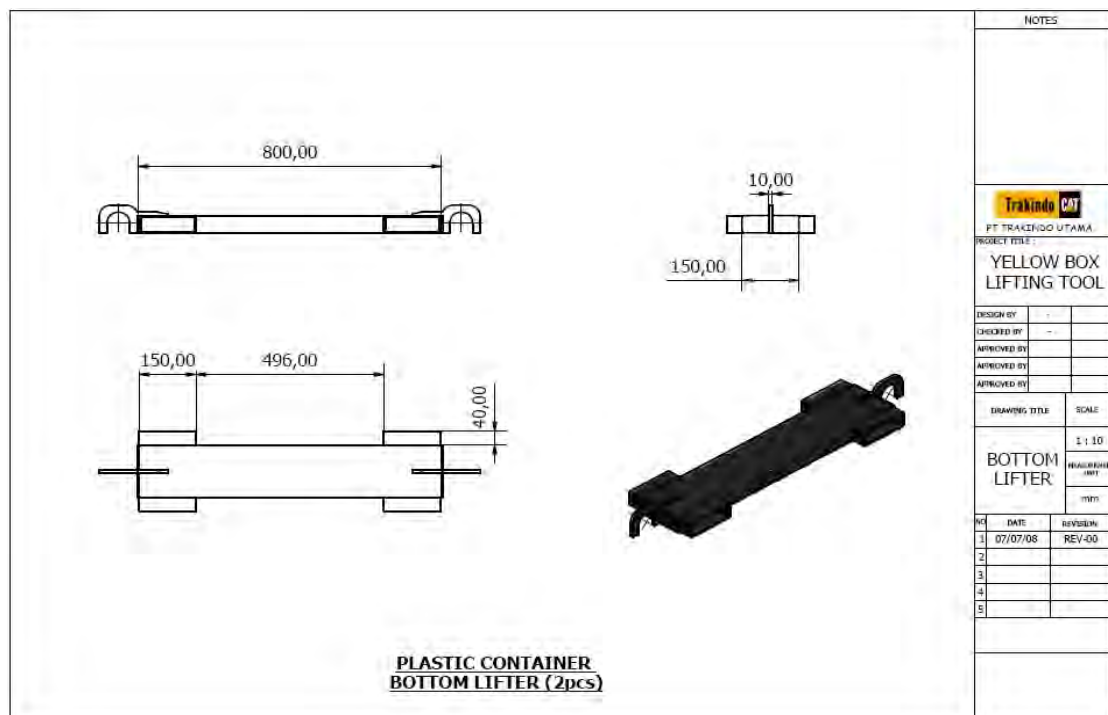


Figure 10: Drawing of bottom portion of lifting tool.

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DATE
2/19/2015CHG
NO
00NUMBER
0911-4.03-1230

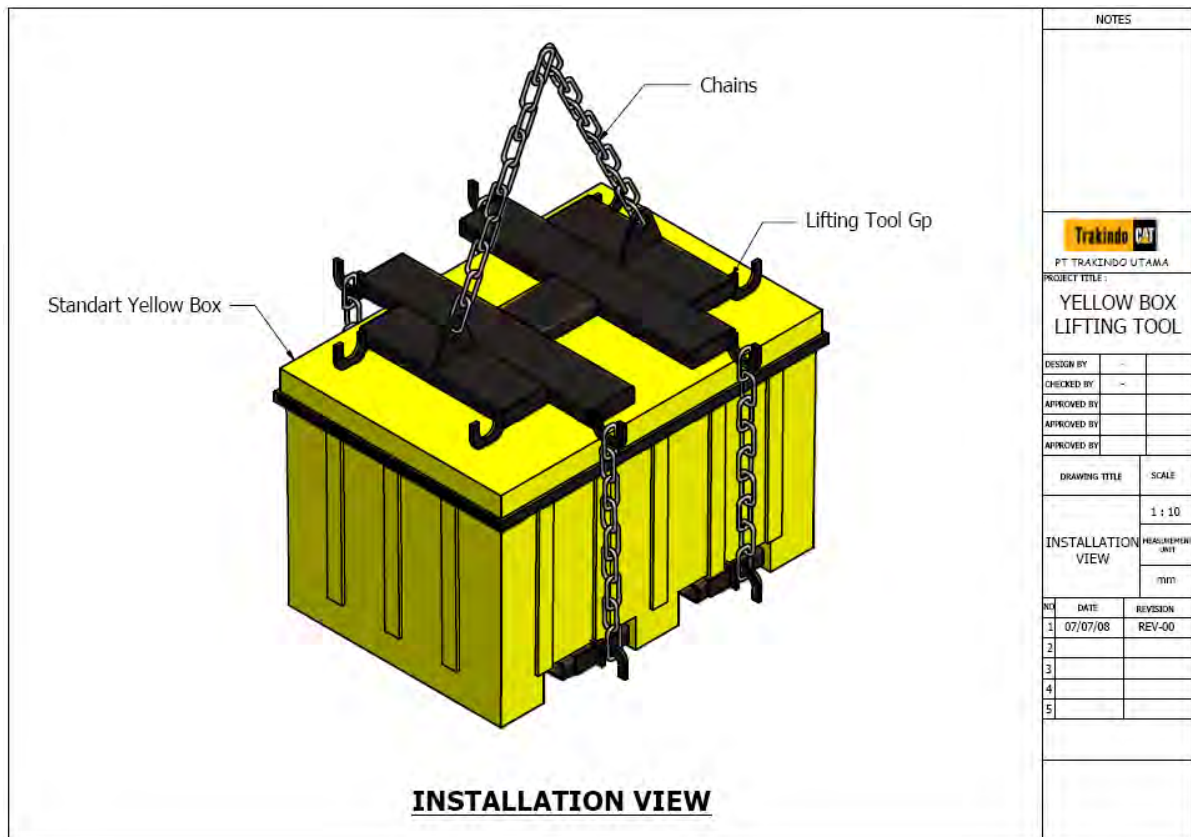


Figure 11: Drawing of lifting tool.

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CHG
NO
00

NUMBER
0911-4.03-1230

Trakindo's Standard Plastic Container (Yellow Box)

- Trakindo utilizes two types yellow storage box with bottom support (type 7099K) and without bottom support (type 7099). These are supplied by PT. Indra raha Nusaplasindo, a local supplier.
- Mostly utilize type 7099K with dimension 1100(L) X 750(W) X 705 (H) mm for double stacking.



PLASTIC CONTAINER

DESCRIPTION

Plastic box to be used to group and protect parts / components from dust. To keep for a while PRA-Reman item that has been inspected before delivered to Caterpillar



TRAKINDO STANDARD COLOUR

Surface colour

Colour : Blue (8000)
: Yellow (7099-7909)
Code : -

SUPPLIER

(LOCAL SUPPLIER)

" PT. INDRA GRAHA NUSAPLASINDO "
" *RABBIT* "

DELIVERY TIME (FRANCO JKT)

3 weeks after LPO submitted by CPD

SPECIFICATION

Recommended Type / Name	: <i>Container plastik rapat / 7099</i>
Dimension- LxWxH (mm)	: 1100 x 750 x 640
PTTU application	: To keep WIP, PRA and Reman items
Recommended Type / Name	: <i>Container plastik rapat / 7099 K</i>
Dimension- LxWxH (mm)	: 1100 x 750 x 705
PTTU application	: To keep WIP, PRA and Reman items
Recommended Type / Name	: <i>Container plastik rapat / 8000</i>
Dimension- LxWxH (mm)	: 660 x 465 x 270
PTTU application	: To keep WIP items

Figure 12: Yellow Box information from PT. Indra raha Nusaplasindo, a local supplier.

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WIP Parts Storage Boxes and Lifting Tool	DATE 2/19/2015	CHG NO 00	NUMBER 0911-4.03-1230

Inside Plastic Container (Yellow Box)

- Parts stored in yellow storage box may vary; it's separated by parts/ component material as shown on the picture:



Figure 13: Storage for New Parts Supply.



Figure 14: Storage for Reusable Parts.



Figure 15: Storage for WIP.



Figure 16: Storage for Reman Parts.



Figure 17: Storage for Reman Parts.



Figure 18: Storage for Tubes.

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WIP Parts Storage Boxes and Lifting Tool	DATE	CHG NO	NUMBER
	2/19/2015	00	0911-4.03-1230

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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WIP Parts Storage Boxes and Lifting Tool	DATE 2/19/2015	CHG NO 00	NUMBER 0911-4.03-1230
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